ACADEMIC THEORY BRIEF · VERSION 2.0

# The Verification Boundary

*Why generation and verification separate into distinct firms as artificial intelligence commoditizes*


## ABSTRACT

This brief contains one primary theoretical claim and a small number of supporting constructs. The claim is that in any market where automated output carries fiduciary, regulatory, safety, or reputational weight, the generator and the verifier of that output separate into distinct economic entities, and the separation is stable under improvements in generator capability. The claim implies a predictable boundary on platform expansion: firms that supply generation can integrate forward into adjacent workflow and interface layers, but cannot credibly integrate into the verification function performed above themselves. The brief states the proposition, positions it against appropriability, ecosystem, modularity, and information-asymmetry literatures, specifies boundary conditions, and offers five falsifiable predictions with explicit refutation criteria. The broader taxonomy from which the proposition is drawn — a ten-layer supply chain of intelligence — is deliberately relegated to appendices and companion documents.



Anand Arivukkarasu — Kellogg MBA; former product leader, Meta (Instagram); VP / Head of Product roles at Ideas2IT, Refersion, GRIN. San Francisco. Written in a personal capacity.


Canonical source: supplychainofai.com/framework · supplychainofai.com/papers

# 1. The proposition

> **Proposition IV — the Verification Boundary**
> Wherever the output of an automated system carries fiduciary, regulatory, safety, or reputational weight, generation and verification separate into distinct economic entities — and that separation is stable under improvements in the generator's capability.

Three terms need fixing before the claim can be evaluated.

- **Generation** is the production of a candidate output: a draft, a diagnosis, a configuration, a line of code, a valuation, a filing.
- **Verification** is an assertion, addressed to a third party, that the output meets a standard. It is not quality control. Quality control is internal and self-interested; verification is external and is consumed by someone who did not commission the work.
- **Economically valuable trust** exists when a party outside the transaction — a regulator, an insurer, an auditor, a court, a board, a counterparty — conditions money or permission on the assertion being credible.

The proposition is therefore not a claim about technical capability. It concedes, in advance, that the generator may become strictly better than the verifier at detecting its own errors. The claim is that this improvement does not dissolve the separation, because the separation is not produced by a capability gap. It is produced by the structure of who is permitted to make a credible assertion about whom.

## The forcing mechanism

1. **Self-referential attestation carries no information.** Following Spence (1973), a signal is informative only when it is costlier to send falsely than truthfully. An attestation issued by the party that produced the output has near-zero differential cost, so a rational receiver discounts it toward zero regardless of its accuracy.
2. **Liability must be assignable to a solvent, separable party.** Bundling generation and verification collapses two claims into one balance sheet, which destroys the insurability of the transaction. Insurers price separation explicitly.
3. **Demand for verification is created by third parties, not by the buyer.** The buyer would often prefer the bundle: it is cheaper. Regulators, auditors and insurers prevent the bundle from forming. Institutional constraints of this kind harden over time rather than eroding (Power, 1997).
4. **Buyers pay a duplication tax to avoid a single-point-of-failure tax.** Where one vendor's error is unrecoverable, procurement selects two vendors. This is observable and priced: it is the two-vendor rule.

## 2. What is genuinely new

Existing theory explains a great deal of what happens in AI markets. It does not, in the author's reading, generate this boundary. The table states the relationship precisely rather than claiming a vacuum.

| Prior work | What it already explains | What Proposition IV adds |
|---|---|---|
| Teece (1986), appropriability and complementary assets | Why innovators capture value only when they hold the complementary assets; why specialized assets shift bargaining power. | Identifies verification as a complementary asset that *cannot be acquired by the innovator without destroying its value*. Standard appropriability logic predicts integration; here integration is self-defeating. |
| Porter (1985), positioning and value chains | Where margin sits in a chain, and how firms defend a position. | Supplies an endogenous rule for where a chain must break into separate firms, rather than treating firm boundaries as a managerial choice. |
| Baldwin & Clark (2000), modularity | How interfaces partition technical systems and where options accumulate. | The boundary here is institutional, not technical. It persists even when the interface cost of integration is zero. |
| Jacobides et al. (2018); Adner (2017), ecosystems | Complementarity, bottlenecks, and multilateral dependence in ecosystems. | Predicts which ecosystem role is structurally non-absorbable by the hub, and gives the condition under which hub expansion stops. |
| Akerlof (1970); Spence (1973); Holmström (1979) | Adverse selection, signalling, and moral hazard under asymmetric information. | Applies the signalling condition to a firm-boundary prediction in a market where the informed party's capability is rising rapidly, and shows why rising capability does not relax the constraint. |
| Coase (1937); Williamson (1985) | Firm boundaries as a function of transaction and governance costs. | Adds a credibility constraint that operates in the opposite direction to transaction cost: integrating reduces cost while destroying the asset's economic function. |

## 3. Boundary conditions

The proposition is intended to be narrow enough to be wrong. It applies where all three conditions hold.

- **Consequence.** Errors in the output impose losses on a party other than the producer, and those losses are not fully recoverable by rework.
- **External conditioning.** Some third party conditions money, permission, or admissibility on the output being certified.
- **Observable standard.** A standard exists, or can be constructed, against which conformity can be asserted.

Where these do not hold — casual content generation, internal drafting, entertainment, exploratory analysis — the proposition makes no prediction, and bundled generation-plus-checking should be expected to dominate. This is why the consumer AI market shows no verification layer and the clinical, financial, legal, and safety-critical markets show a durable one.

## 4. Falsifiable predictions

Each prediction states an observable and the observation that would refute it. All are stated over a five-year horizon from publication.

| # | Prediction | Refuted if |
|---|---|---|
| P1 | No frontier model provider will issue binding third-party certification of its own outputs that a regulated buyer accepts in place of an independent verifier. | A model provider's self-issued attestation is accepted by an auditor, insurer, or regulator as a substitute for independent verification in a regulated workflow. |
| P2 | In regulated verticals, verification vendors will show lower revenue churn and longer contract duration than generation vendors serving the same buyers. | Verification vendors churn at or above generation vendors over a comparable cohort. |
| P3 | Improvements in generator accuracy will not reduce spend on independent verification; verification spend will be flat or rising as generation improves. | A sustained decline in verification spend that tracks generator accuracy improvements, controlling for total AI spend. |
| P4 | Platform expansion from generation will proceed into adjacent execution, orchestration and interface functions, but will stop at verification performed above the platform itself. | A generation platform successfully absorbs the verification function above it and retains regulated-buyer acceptance for more than two audit cycles. |
| P5 | Where a verification requirement is newly imposed by regulation, an independent verification vendor will emerge or be acquired-and-ring-fenced within two years, rather than the function being absorbed by incumbent generators. | Newly regulated domains routinely resolve into single-vendor generation-plus-verification without structural separation. |

### What would refute the proposition as a whole

A single, well-documented market in which all three boundary conditions hold and the generator's own attestation is durably accepted by the conditioning third party would force an amendment. Capability arguments alone would not: the proposition explicitly predicts that the generator becomes more accurate and the separation persists anyway.

## 5. Supporting constructs, deliberately deferred

The proposition sits inside a larger descriptive framework, the Supply Chain of Intelligence, which partitions generative AI markets into ten layers from physical resources to compounding memory, and states three further structural regularities about where value migrates. That taxonomy is an instrument for locating firms; it is not part of the theoretical claim, and evaluating the claim does not require accepting it. It is documented in the companion working paper (Appendix A) and in the practitioner guide, and is available in full at the canonical source.

# References


Akerlof, G. A. (1970). The market for “lemons”: quality uncertainty and the market mechanism. *Quarterly Journal of Economics*, 84(3), 488–500.

Adner, R. (2017). Ecosystem as structure: an actionable construct for strategy. *Journal of Management*, 43(1), 39–58.

Arrow, K. J. (1962). Economic welfare and the allocation of resources for invention. In *The Rate and Direction of Inventive Activity*. Princeton University Press.

Baldwin, C. Y., & Clark, K. B. (2000). *Design Rules: The Power of Modularity*. MIT Press.

Barney, J. (1991). Firm resources and sustained competitive advantage. *Journal of Management*, 17(1), 99–120.

Christensen, C. M. (1997). *The Innovator’s Dilemma*. Harvard Business School Press.

Coase, R. H. (1937). The nature of the firm. *Economica*, 4(16), 386–405.

Holmström, B. (1979). Moral hazard and observability. *Bell Journal of Economics*, 10(1), 74–91.

Jacobides, M. G., Cennamo, C., & Gawer, A. (2018). Towards a theory of ecosystems. *Strategic Management Journal*, 39(8), 2255–2276.

Porter, M. E. (1985). *Competitive Advantage: Creating and Sustaining Superior Performance*. Free Press.

Power, M. (1997). *The Audit Society: Rituals of Verification*. Oxford University Press.

Rochet, J.-C., & Tirole, J. (2003). Platform competition in two-sided markets. *Journal of the European Economic Association*, 1(4), 990–1029.

Shapiro, C., & Varian, H. R. (1998). *Information Rules*. Harvard Business School Press.

Spence, M. (1973). Job market signaling. *Quarterly Journal of Economics*, 87(3), 355–374.

Teece, D. J. (1986). Profiting from technological innovation. *Research Policy*, 15(6), 285–305.

Williamson, O. E. (1985). *The Economic Institutions of Capitalism*. Free Press.


## How to cite


Arivukkarasu, A. (2026). *Supply Chain of Intelligence: where competitive advantage accumulates in AI markets* (Version 2.0). https://supplychainofai.com/papers

BibTeX: @misc{arivukkarasu2026scoi, author = {Arivukkarasu, Anand}, title = {Supply Chain of Intelligence}, year = {2026}, note = {Version 2.0}, url = {https://supplychainofai.com/papers}}